

Hybrid Digital Twin for Real-Time Thermal Runaway Detection in Lithium-Ion Batteries

Overview

This project presents a Hybrid Digital Twin framework for early detection of thermal runaway in lithium-ion batteries. It combines a physics-based thermal model with an LSTM deep learning model using an adaptive fusion algorithm. The implementation is simulation-based and validated using a benchmark battery dataset in Google Colab.

Key Features

Physics-Based Digital Twin
LSTM Temperature Prediction
Adaptive Fusion Algorithm
Battery Risk Assessment
Thermal Runaway Detection
Early Warning System
Real-Time Dataset Replay Simulation
Performance Evaluation

Architecture

Battery Dataset → Data Preprocessing → Physics Digital Twin + LSTM Model → Adaptive Fusion → Hybrid Prediction → Battery Risk Assessment → Thermal Runaway Warning → Performance Evaluation

Workflow

1. Load Dataset
2. Data Cleaning & EDA
3. Physics Model
4. LSTM Model
5. Adaptive Fusion
6. Risk Assessment
7. Thermal Runaway Detection
8. Real-Time Simulation
9. Performance Evaluation

Technologies

Python, Google Colab, Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow/Keras

Physics-Based Model

Uses Joule's Law ($Q = I^2R$) to estimate heat generation and battery temperature.

AI Model

An LSTM network predicts future battery temperature from historical battery sensor readings.

Adaptive Fusion

Combines physics and AI predictions using adaptive weights based on battery operating conditions.

Outputs

Hybrid Temperature Prediction, Battery Risk Score, Battery Status, Thermal Runaway Warning, Performance Metrics.

Applications

Electric Vehicles, Battery Management Systems, Energy Storage Systems, Smart Charging Stations.

Disclaimer

This is a simulation-based Digital Twin validated using a benchmark battery dataset. Future work includes real hardware integration with a Battery Management System.

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